WORKED SOLUTION

STEP I 2018, Question 4

Pure • Sketching an integral function

STEP I 2018, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q4

The key idea

Everything in f is built from $\ln x$, and the awkward factor $\frac{1}{x}$ is exactly the derivative of $\ln x$. So the substitution $u = \ln x$ turns the integral into the integral of a simple rational function of u. In terms of u, replacing x by x^{-1} just changes the sign of u, so the symmetry comes down to the answer being an even function of u.

Solution

Throughout, write $u = \ln x$, so $u \neq 0$ because $x \neq 1$, and $u > 0$ exactly when $x > 1$.

The zeros of f and f′

Let

$$g(u) = \frac{(1-u^2)^2}{u} = \frac{1}{u} - 2u + u^3, \qquad \text{so that} \qquad \mathrm{f}(x) = \frac{g(\ln x)}{x}.$$

When $(\ln x)^2 = 1$, the factor $(1-u^2)^2$ is zero, so $\mathrm{f}(x) = 0$. By the quotient rule and the chain rule,

$$\mathrm{f}'(x) = \frac{g'(u) \cdot \frac{1}{x} \cdot x - g(u)}{x^2} = \frac{g'(u) - g(u)}{x^2}.$$

We already have $g(\pm 1) = 0$. Also, because $(1-u^2)^2 = (1-u)^2(1+u)^2$ has a *repeated* factor at $u = \pm 1$, its derivative $-4u(1-u^2)$ vanishes there too, and so

$$g'(u) = \frac{-4u(1-u^2) \cdot u - (1-u^2)^2}{u^2} = 0 \quad \text{at } u = \pm 1.$$

Hence $\mathrm{f}'(x) = 0$ as well whenever $(\ln x)^2 = 1$, that is, at $x = e$ and $x = e^{-1}$.

Part (i)

Substitute $u = \ln t$, so $\mathrm{d}u = \frac{1}{t}\,\mathrm{d}t$. Then

$$\int \mathrm{f}(t)\,\mathrm{d}t = \int \frac{(1-u^2)^2}{u}\,\mathrm{d}u = \int \left(\frac{1}{u} - 2u + u^3\right)\mathrm{d}u = \ln|u| - u^2 + \tfrac{1}{4}u^4 + C.$$

Define $G(u) = \ln|u| - u^2 + \frac{1}{4}u^4$. The two pieces of F never let the range of integration cross $t = 1$, so u keeps one sign on each and the antiderivative is valid there.

For $x > 1$, the lower limit $t = e$ gives $u = 1$ and $G(1) = 0 - 1 + \frac{1}{4} = -\frac{3}{4}$. For $0 < x < 1$, the lower limit $t = e^{-1}$ gives $u = -1$ and $G(-1) = -\frac{3}{4}$ too. So both cases give the same formula:

$$\boxed{\mathrm{F}(x) = \ln|\ln x| - (\ln x)^2 + \tfrac{1}{4}(\ln x)^4 + \tfrac{3}{4} \qquad (x > 0,\ x \neq 1).}$$

Now $\ln(x^{-1}) = -\ln x$. Every appearance of $\ln x$ above is inside $|\cdot|$ or raised to an even power, so

$$\mathrm{F}(x^{-1}) = \ln|-\ln x| - (-\ln x)^2 + \tfrac{1}{4}(-\ln x)^4 + \tfrac{3}{4} = \mathrm{F}(x),$$

as required. (The constant $\frac{3}{4}$ is the same on both sides only because the two lower limits e and e^{-1} are themselves a reciprocal pair.)

Part (ii)

? **Ask yourself** Where is F increasing, and what happens next to $x = 1$?

Gradient. $\mathrm{F}'(x) = \mathrm{f}(x)$. The factor $(1-u^2)^2$ is never negative and $x > 0$, so $\mathrm{f}(x)$ has the sign of $u = \ln x$. So F is decreasing on $(0,1)$ and increasing on $(1,\infty)$. Its only stationary points are at $x = e^{-1}$ and $x = e$, where $\mathrm{F} = 0$ by definition.

These are not turning points. Near $x = e$ the gradient is positive on both sides, and near $x = e^{-1}$ it is negative on both sides. Since $\mathrm{f}' = \mathrm{F}''$ is also zero there, they are *stationary points of inflection*. It follows that

$$\mathrm{F}(x) > 0 \text{ for } 0 < x < e^{-1} \text{ and } x > e, \qquad \mathrm{F}(x) < 0 \text{ for } e^{-1} < x < 1 \text{ and } 1 < x < e.$$

Behaviour at the ends.

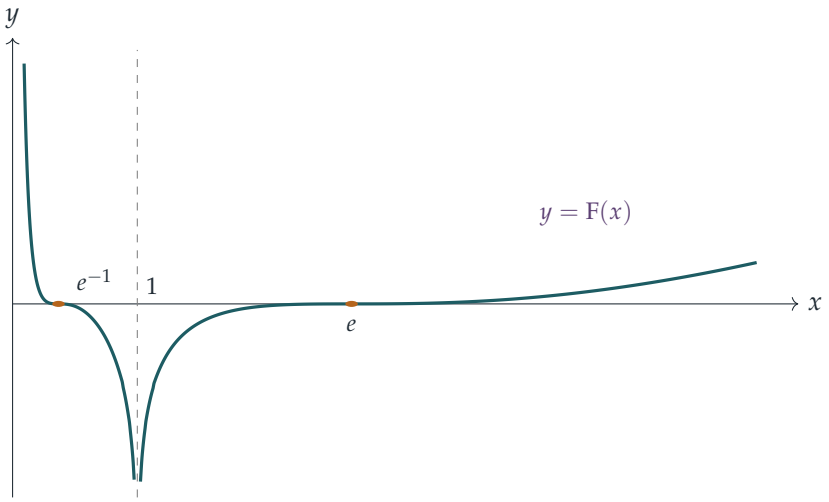
- As $x \to 1$, $u \to 0$ and $\ln|u| \to -\infty$, while the other terms tend to $\frac{3}{4}$. So $\mathrm{F}(x) \to -\infty$ from both sides, and $x = 1$ is a vertical asymptote.
- As $x \to \infty$, $u \to \infty$ and the $\frac{1}{4}u^4$ term dominates, so $\mathrm{F}(x) \to \infty$. It does so very slowly: the gradient $\mathrm{f}(x) \approx u^3/x \to 0$.
- As $x \to 0^+$, $u \to -\infty$, so again $\mathrm{F}(x) \to \infty$, by the symmetry. Here it does so steeply: the gradient $\mathrm{f}(x) \approx u^3/x \to -\infty$. The curve does not meet the y-axis.

Concavity (for a more careful sketch). Using $\mathrm{F}'' = \mathrm{f}' = (g' - g)/x^2$ and simplifying,

$$\mathrm{F}''(x) = -\frac{(u^2-1)(u^3 - 3u^2 - u - 1)}{u^2x^2}.$$

The cubic $u^3 - 3u^2 - u - 1$ has a single real root, at $u \approx 3.38$ (its local maximum value, at $u = 1 - \frac{2}{\sqrt{3}}$, is negative). It is negative for every smaller u. So the curve is concave up on $(0, e^{-1})$ and concave down on $(e^{-1}, 1)$ and on $(1, e)$. It is concave up from e until a further inflection at $x \approx 29.5$, and concave down after that. That last inflection is far to the right and needn't appear on a sketch.

The symmetry $\mathrm{F}(x^{-1}) = \mathrm{F}(x)$ says the height at x matches the height at $1/x$. For example $\mathrm{F}(2) = \mathrm{F}(\frac{1}{2}) \approx -0.04$ and $\mathrm{F}(5) = \mathrm{F}(\frac{1}{5}) \approx 0.31$. So the part of the curve on $(1, \infty)$ is the part on $(0, 1)$ squashed horizontally towards the asymptote.



The curve comes down steeply from $+\infty$ near the y-axis. It levels off to touch the x-axis at a stationary inflection at $(e^{-1}, 0)$, then plunges to $-\infty$ at the asymptote $x = 1$. On the right it climbs back from $-\infty$, flattens to a second stationary inflection at $(e, 0)$, and then rises slowly to $+\infty$.

Checks

- Differentiate the boxed formula: $\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln|u| - u^2 + \frac{1}{4}u^4\right) = \frac{1}{x}\left(\frac{1}{u} - 2u + u^3\right) = \mathrm{f}(x)$, and the formula gives 0 at $x = e$ and at $x = e^{-1}$.
- Numerically, $\int_e^5 \mathrm{f}(t)\,\mathrm{d}t \approx 0.3130$ and $\int_{1/e}^{1/5} \mathrm{f}(t)\,\mathrm{d}t \approx 0.3130$, matching the formula and the symmetry.
- The signs fit: on $(1, e)$ the function increases to 0, so it must be negative there, and indeed $\mathrm{F}(2) \approx -0.039$.

Common slips

- Writing $\ln(\ln x)$ for $0 < x < 1$, where $\ln x < 0$. The antiderivative of $\frac{1}{u}$ is $\ln|u|$, and the modulus is exactly what makes the symmetry work.

- Drawing turning points at $x = e^{\pm 1}$. Because f has a squared factor, it does not change sign there, so these are stationary points of inflection.
- Missing the vertical asymptote at $x = 1$, or joining the two branches across it.

Beyond STEP

F is defined by an integral, and the fundamental theorem of calculus says $\mathrm{F}' = \mathrm{f}$ on each interval where f is continuous. That is why the two pieces need separate base points: f blows up at $x = 1$, so no single integral can be taken across it. The symmetry $x \mapsto x^{-1}$, which becomes $u \mapsto -u$ under $u = \ln x$, is a small example of turning a multiplicative symmetry into an additive one, an idea that runs through analysis. For a careful account of the fundamental theorem and improper integrals, see Abbott, *Understanding Analysis*.

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